# Foundation Models Baseline

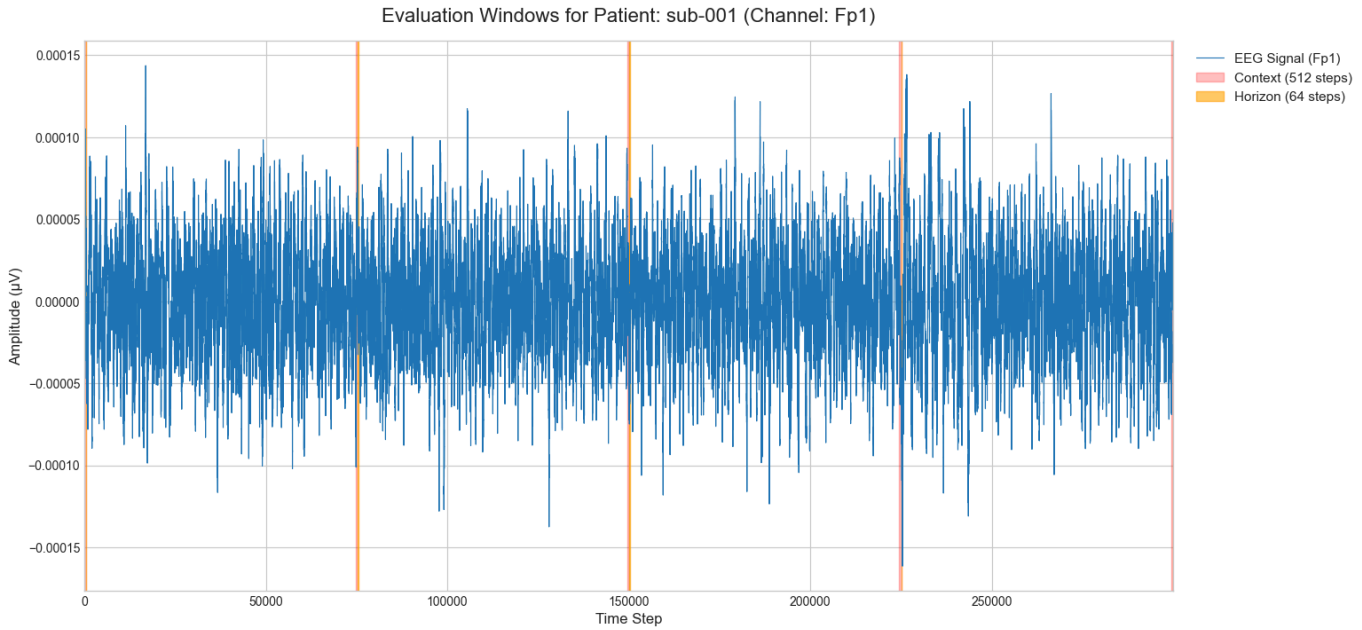
## Input:

- Preprocessed EEG signals from 4 channels: Fp1, Fp2, P3, P4
- `FS` - 500 Hz (Original sampling frequency)
- `CONTEXT_LENGTH` - 512 (~1.02 s)
- `HORIZON_LENGTH` - 64 (~0.13 s)
- Window selection - 5 windows evenly spaced across the signal - np.linspace(0, max_start, 5)

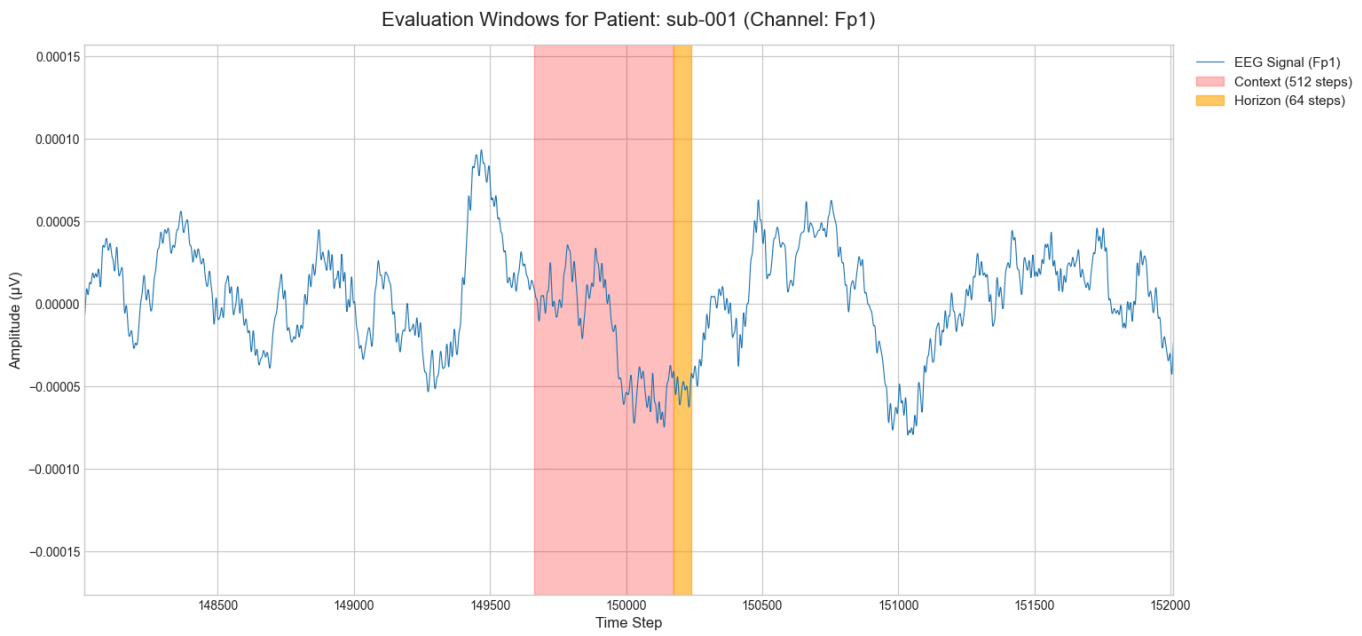
## Evaluation:

- MSE on 64 samples -> mean from 5 windows

### Windows over signal:




### Singla window zoom:




## Used models, specification:

| Model | Architecture / Size | Prediction Mode | Number of Samples (for Median) | Additional info |
|---|---|---|---|---|
| **Chronos** | T5-Base (NLP) | Probabilistic | 20 | - |
| **TimesFM** | Patching (200M) | Point | N/A | - |
| **Moirai** | Universal Transformer (Base) | Probabilistic | 20 | Artificial sampling frequency forced to 1 second (`freq="S"`). |
| **Lag-Llama** | Llama-based | Probabilistic | 20 | Linear scaling of Rotary Position Embeddings applied (RoPE scaling factor: `(512+64)/32`) to handle the extended context window. |
| **TimeGPT** | Zero-Shot Cloud API | Point | N/A | Parameter `freq="S"`. Forced autoregressive mode (API warning) due to the horizon length (64). |
| **Sundial** | Causal LM (Base-128M) | Probabilistic | 20 | `transformers==4.40.1` required to turn on. |
| **ViTime** | Vision Transformer (ViT) | Probabilistic | 20 | Numerical conversion into 2D binary images. |
| **TimeFound** | Encoder-Decoder (Base- | Point | N/A | Independent normalization (`StandardScaler`, z-score) for each |

| Model | Architecture / Size | Prediction Mode | Number of Samples (for Median) | Additional info |
|---|---|---|---|---|
| | 200M) | | | window separately before input, and inverse transformation at the output. |

## Results:

Table 1: Overall Performance (Mean Squared Error)

| Model | Overall Mean MSE |
|---|---|
| Timegpt | 4.61399e-10 |
| Chronos | 4.73524e-10 |
| Sundial | 4.7541e-10 |
| Moirai | 2.31305e-08 |
| Vitime | 5.22668e-08 |
| Timesfm | 9.25312e-07 |

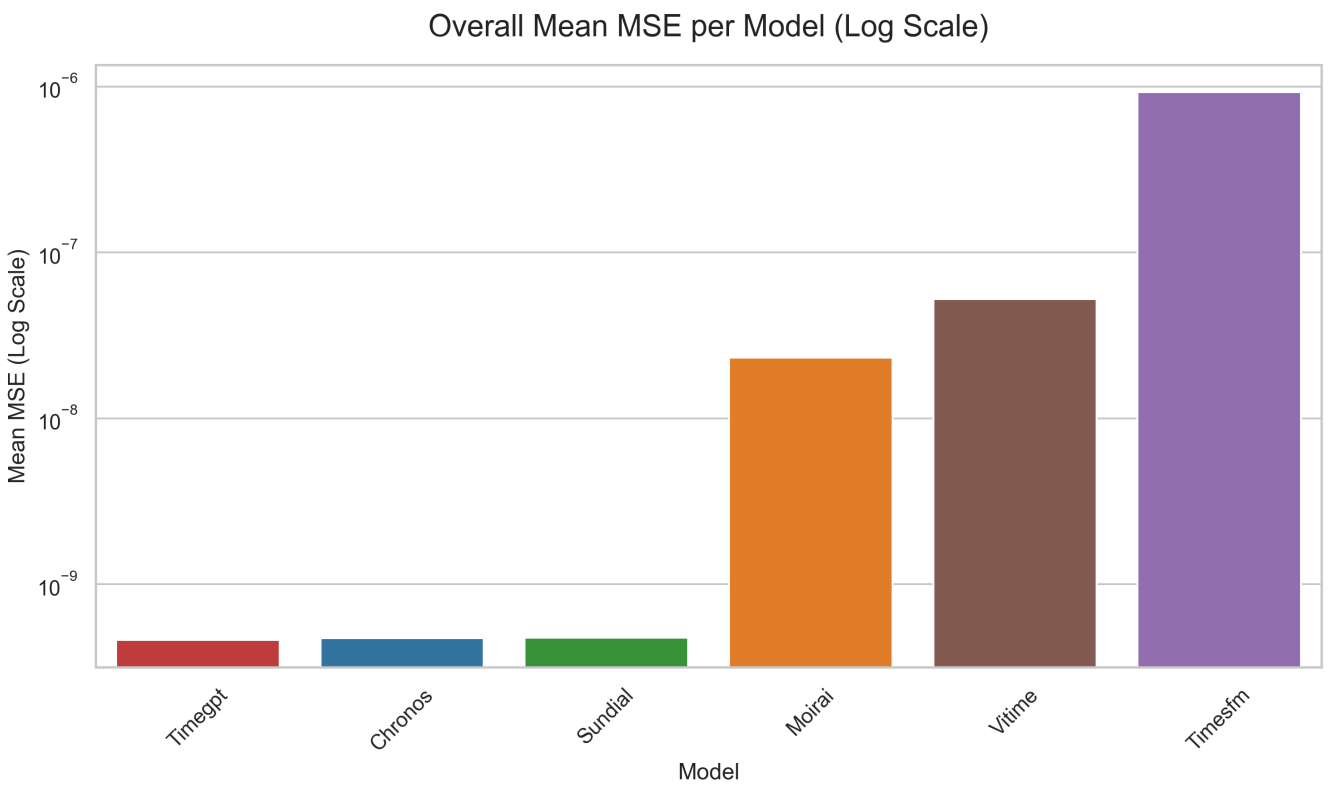



Table 3: Performance by Patient Group (Mean Squared Error)

| Model | A | C | F | Average |
|---|---|---|---|---|
| Timegpt | 4.70058e-10 | 4.64978e-10 | 4.43333e-10 | 4.59456e-10 |

| Model | A | C | F | Average |
|---|---|---|---|---|
| Chronos | 5.19075e-10 | 4.35622e-10 | 4.50014e-10 | 4.68237e-10 |
| Sundial | 5.10719e-10 | 4.52262e-10 | 4.49332e-10 | 4.70771e-10 |
| Moirai | 3.7279e-08 | 1.3231e-08 | 1.3467e-08 | 2.13257e-08 |
| Vitime | 5.25987e-08 | 5.1552e-08 | 5.26484e-08 | 5.22664e-08 |
| Timesfm | 2.22153e-06 | 4.35667e-10 | 6.25908e-08 | 7.6152e-07 |

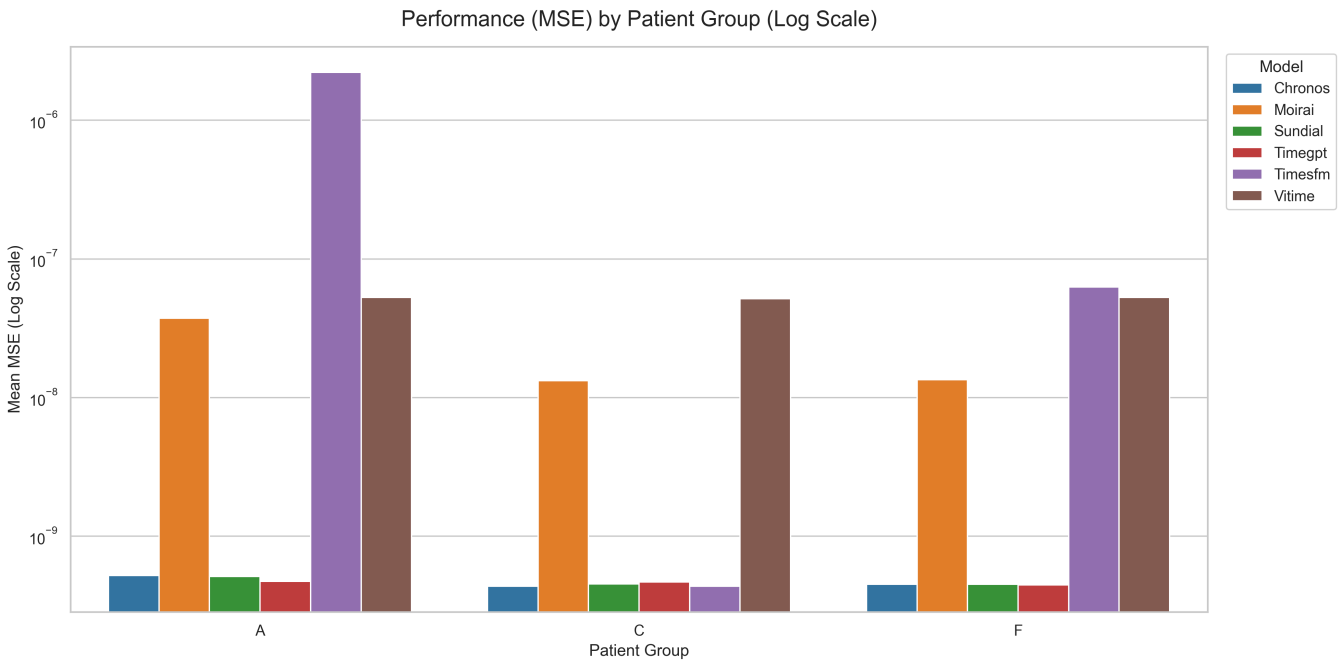
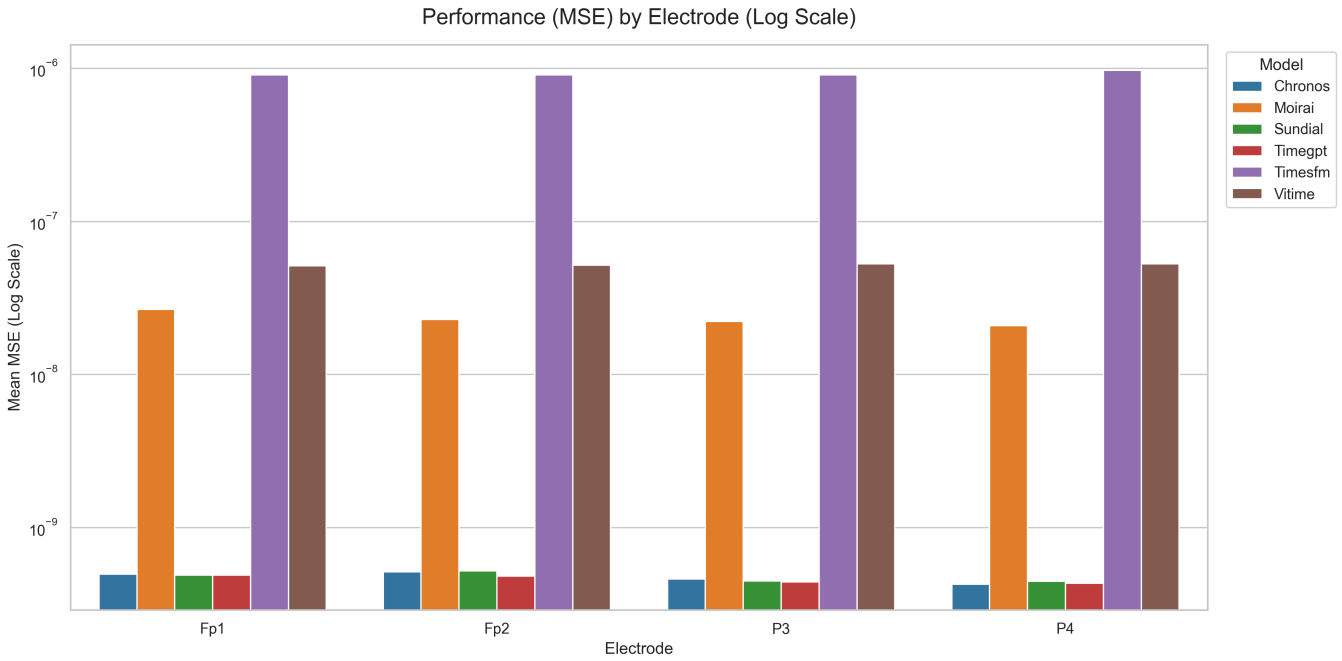



Table 2: Performance by Electrode (Mean Squared Error)

| Model | Fp1 | Fp2 | P3 | P4 | Average |
|---|---|---|---|---|---|
| Timegpt | 4.89381e-10 | 4.81693e-10 | 4.41102e-10 | 4.33419e-10 | 4.61399e-10 |
| Chronos | 4.94979e-10 | 5.12373e-10 | 4.60068e-10 | 4.26675e-10 | 4.73524e-10 |
| Sundial | 4.88827e-10 | 5.206e-10 | 4.47678e-10 | 4.44537e-10 | 4.7541e-10 |
| Moirai | 2.66403e-08 | 2.28446e-08 | 2.22077e-08 | 2.08295e-08 | 2.31305e-08 |
| Vitime | 5.14331e-08 | 5.18469e-08 | 5.29789e-08 | 5.28081e-08 | 5.22668e-08 |
| Timesfm | 9.10171e-07 | 9.08812e-07 | 9.08597e-07 | 9.73669e-07 | 9.25312e-07 |


Performance (MSE) by Electrode (Log Scale)
Mean MSE (Log Scale)
$10^{-6}$
$10^{-7}$
$10^{-8}$
$10^{-9}$
Fp1
Fp2
P3
P4
Electrode
Model
Chronos
Moirai
Sundial
Timegpt
Timesfm
Vitime